

---

# Beyond Perplexity: Evaluating LLM Token Distributions with Proper Scoring Rules

---

Zhicheng You

Sagarima Datta

## Abstract

The evaluation of language models is dominated by perplexity, the exponentiated mean log-score, which assesses predictive distributions exclusively through the Bregman geometry of the Kullback–Leibler divergence. We evaluate three open autoregressive language models (GPT-2, GPT-2-Medium, OPT-125M) on WikiText-103 and the Penn Treebank under four strictly proper scoring rules—the log-score, the quadratic (Brier) score, the energy score, and the Gaussian-kernel (MMD) score—each of which, by the Savage representation, evaluates forecasts through the Bregman divergence of a distinct convex generator. Across 20,000 scored token positions per model–corpus pair we report four findings. First, the rules do not rank the models identically: the energy score prefers OPT-125M while the remaining rules unanimously prefer GPT-2-Medium, and a nonparametric bootstrap reproduces this disagreement in every resample. Second, the penalty each rule assigns to rare tokens scales with the boundary curvature of its generator: from the most- to the least-frequent decile the log-score grows  $2.05\times$ , against  $1.22\times$  for the bounded quadratic score. Third, the log-to-energy penalty ratio widens by 47% from low- to high-entropy predictions, as the locality of the Kullback–Leibler divergence predicts. Fourth, a centroid analysis of more than 13,000 misprediction events shows that the energy ranking tracks the distance between the predictive barycenter and the realized token in embedding space—the quantity the energy divergence is constructed to measure. Together, the experiments constitute an empirical verification of the Bregman account of scoring-rule disagreement and motivate reporting complementary proper scoring rules alongside perplexity. Code: <https://github.com/sagarima-datta/beyond-perplexity>.

## 1 Introduction

At every generation step an autoregressive language model produces a complete probability distribution over its vocabulary. The prevailing evaluation practice reduces this object to a single statistic, perplexity, computed from the log-score  $-\log p(y)$  of the realized token  $y$ . The choice is historically motivated (Jelinek et al., 1977) rather than derived from first principles: the log-score is one member of the family of *strictly proper scoring rules* (Gneiting and Raftery, 2007), and each member of that family evaluates forecasts through a different divergence geometry. Reliance on perplexity alone therefore amounts to an implicit assumption that the Kullback–Leibler (KL) geometry is appropriate for all model comparisons. The present work examines that assumption empirically.

We score three publicly available causal language models under four strictly proper scoring rules on two corpora, asking three questions. **(Q1)** Do the rules induce the same model ranking? **(Q2)** Where they disagree, is the disagreement statistically robust, and is it predicted by the geometry of the rules’ Bregman generators? **(Q3)** Can the mechanism of a disagreement be located in the structure of the predictive distributions themselves?

**Contributions.** The paper makes three contributions.

1. A systematic comparison of four strictly proper scoring rules (log, quadratic, energy, kernel) over 3 models  $\times$  2 corpora  $\times$  20,000 token positions, with per-token score caches permitting token-level concordance analysis (Sections 4–6).
2. A bootstrap certification that the observed inter-rule ranking disagreement is not attributable to sampling variability (Section 7).
3. An experiment-by-experiment verification that the observed score behavior is the behavior the Savage–Bregman representation requires: tail penalties order with generator curvature (Section 5), uncertainty penalties with locality (Section 6), and the energy ranking coincides with the ordering of predictive-barycenter distances in embedding space (Section 8).

## 2 Theoretical background

### 2.1 Proper scoring rules

Let  $\mathcal{V} = \{1, \dots, V\}$  denote the vocabulary and  $\Delta^{V-1}$  the probability simplex over  $\mathcal{V}$ . A (negatively oriented) scoring rule is a function  $S : \Delta^{V-1} \times \mathcal{V} \rightarrow \mathbb{R}$  assigning loss  $S(P, y)$  to the forecast  $P$  upon realization of  $y$ . Writing  $S(P, Q) = \mathbb{E}_{y \sim Q} S(P, y)$  for the expected loss under data distribution  $Q$ , the rule is *proper* if  $S(Q, Q) \leq S(P, Q)$  for all  $P, Q \in \Delta^{V-1}$ , and *strictly proper* if equality implies  $P = Q$ . Strict properness is the minimal requirement for honest evaluation: it renders truthful reporting of beliefs the unique minimizer of expected loss.

### 2.2 The Bregman representation theorem

The structural result on which our analysis rests is the Savage representation (Gneiting and Raftery, 2007): a regular scoring rule  $S$  is (strictly) proper if and only if there exists a (strictly) convex function  $\varphi : \Delta^{V-1} \rightarrow \mathbb{R}$ —the negative generalized entropy of the rule, recoverable as  $\varphi(P) = -S(P, P)$ —such that

$$S(P, y) = -\varphi(P) - \langle \nabla \varphi(P), e_y - P \rangle, \quad (1)$$

where  $e_y$  denotes the vertex of the simplex at  $y$ . The score of an outcome is thus the negated evaluation, at  $e_y$ , of the supporting hyperplane of  $\varphi$  drawn at the forecast. Taking expectations of Equation (1) under  $y \sim Q$  and subtracting the oracle score yields the divergence form

$$S(P, Q) - S(Q, Q) = d_\varphi(Q, P) := \varphi(Q) - \varphi(P) - \langle \nabla \varphi(P), Q - P \rangle, \quad (2)$$

the Bregman divergence generated by  $\varphi$ , i.e., the gap between  $\varphi$  at  $Q$  and the tangent of  $\varphi$  constructed at  $P$ . Three consequences of Equation (2) organize the experiments that follow.

First, *properness is convexity*. The expected regret on the left of Equation (2) is nonnegative precisely because a convex function dominates its tangents, and vanishes uniquely at  $P = Q$  precisely when  $\varphi$  is strictly convex; the properness of a rule is constituted by its geometry.

Second, *each rule is a choice of geometry*. The generator  $\varphi$  endows the simplex with a curvature, and  $d_\varphi$  measures forecast error along it. By Equation (2), “model  $A$  outperforms model  $B$  under rule  $S$ ” is the statement that  $A$ ’s predictive distributions lie closer to the data distribution in the geometry of  $\varphi_S$ . Table 1 records the generators and divergences of the four rules evaluated here. The curvature of the log-score’s generator,  $\nabla^2 \varphi = \text{diag}(1/p_v)$ , diverges at the boundary of the simplex; the quadratic generator has constant Hessian  $2I$ ; the energy and kernel generators inherit their curvature from the metric structure of an embedding space and are consequently the only rules in the suite for which the notion of *similarity between tokens* is defined. These curvature properties yield concrete, falsifiable predictions about score behavior—predictions the experiments below test.

Third, *inter-rule disagreement is informative*. If two strictly proper rules rank models differently, Equation (2) implies the models’ distributions stand at different relative distances from the data distribution under two distinct curvatures. Throughout Sections 4–8 we accordingly treat each disagreement as a measurement to be explained, asking in each case whether the observed pattern is the one the relevant generators require.

### 2.3 The four rules

All rules below are strictly proper and negatively oriented (lower is better). We write  $p_v$  for the forecast probability of token  $v$  and  $X, X' \stackrel{\text{iid}}{\sim} P$ .

Table 1: Savage–Bregman generators of the four evaluated rules. Here  $X, X' \stackrel{\text{iid}}{\sim} P$ ,  $e(\cdot)$  denotes the embedding map, and  $k$  is the Gaussian kernel. Each generator is recovered as  $\varphi(P) = -S(P, P)$ ; strict convexity of  $\varphi$  is equivalent to strict properness of the rule.

| Rule      | Generator $\varphi(P)$                           | Divergence $d_\varphi$              |
|-----------|--------------------------------------------------|-------------------------------------|
| Log       | $\sum_v p_v \log p_v$ (negative Shannon entropy) | Kullback–Leibler                    |
| Quadratic | $\ p\ _2^2 - 1$                                  | squared Euclidean on $\Delta^{V-1}$ |
| Energy    | $-\frac{1}{2} \mathbb{E} \ e(X) - e(X')\ $       | energy distance (via $e$ )          |
| Kernel    | $\mathbb{E} k(e(X), e(X')) - 1$                  | squared MMD (via $e, k$ )           |

**Log-score.**  $S_{\log}(P, y) = -\log p_y$ , with KL divergence as its Bregman divergence. The rule is *local*—it depends on  $P$  only through the coordinate  $p_y$ , and is in fact the only smooth, local, strictly proper rule—but locality entails that the score is unbounded and insensitive to the placement of the remaining  $1 - p_y$  probability mass.

**Quadratic (Brier) score.**  $S_{\text{quad}}(P, y) = \|p - e_y\|_2^2$ , with squared Euclidean distance as its divergence. The rule is bounded on  $[0, 2]$  and global: mass misplaced anywhere on the simplex is penalized, though no metric structure on the vocabulary enters.

**Energy score.** Given a feature map  $v \mapsto e(v) \in \mathbb{R}^d$ —here, each model’s input-embedding matrix  $E \in \mathbb{R}^{V \times d}$ —the energy score (Gneiting and Raftery, 2007; Székely and Rizzo, 2013) is

$$S_E(P, y) = \mathbb{E} \|e(X) - e(y)\| - \frac{1}{2} \mathbb{E} \|e(X) - e(X')\|. \quad (3)$$

Its divergence is the energy distance between  $P$  and the point mass  $\delta_y$ , computed through  $e(\cdot)$ . The rule is distance-sensitive: errors concentrated in the embedding neighborhood of the target are penalized less than errors of equal probability dispersed far from it.

**Kernel (MMD) score.** For the Gaussian kernel  $k(x, x') = \exp(-\|x - x'\|^2 / 2\sigma^2)$ ,

$$S_K(P, y) = k(e(y), e(y)) - 2 \mathbb{E} k(e(X), e(y)) + \mathbb{E} k(e(X), e(X')), \quad (4)$$

the squared maximum mean discrepancy between  $P$  and  $\delta_y$  in the reproducing-kernel Hilbert space of  $k$  (Gretton et al., 2012; Steinwart and Ziegel, 2021). The exponential decay of the kernel beyond its bandwidth  $\sigma$  gives the rule a locality window: mass at distances much greater than  $\sigma$  from the target contributes to the score only through the self-similarity term. We set  $\sigma$  per model by the median heuristic (the median pairwise distance between vocabulary embeddings), the standard choice in the MMD literature.

### 3 Experimental setup

**Models.** GPT-2 (117M parameters), GPT-2-Medium (345M) (Radford et al., 2019), and OPT-125M (Zhang et al., 2022), obtained from the Hugging Face Hub and evaluated in single precision on CPU. The GPT-2 variants share a byte-level BPE tokenizer (50,257 types); OPT employs a closely related vocabulary (50,265 types).

**Corpora.** The WikiText-103 test split (encyclopedic prose) and the Penn Treebank as distributed through NLTK (newswire). Documents are tokenized with each model’s own tokenizer, truncated to 512 tokens, and scored autoregressively until 20,000 positions are accumulated per (model, corpus) cell.

**Estimation.** The log and quadratic scores are computed in closed form from the softmax vector. The energy and kernel scores are estimated by paired Monte Carlo sampling: at each position two independent samples of size  $m = 200$  are drawn from  $P$  and substituted into the unbiased plug-in forms of Equations (3) and (4) (Appendix A). Kernel bandwidths obtained by the median heuristic are  $\sigma_0 = 4.82$  (GPT-2), 4.37 (GPT-2-Medium), and 1.97 (OPT-125M). Per-token scores for every cell are cached, so all downstream analyses operate on identical score arrays.

Table 2: Mean scores over 20,000 token positions (lower is better). Boldface marks the best model under each rule and corpus.

| Corpus       | Model        | Log          | Quadratic    | Energy       | Kernel       |
|--------------|--------------|--------------|--------------|--------------|--------------|
| WikiText-103 | GPT-2        | 3.917        | 0.807        | 1.466        | 0.205        |
|              | GPT-2-Medium | <b>3.591</b> | <b>0.774</b> | 1.223        | <b>0.183</b> |
|              | OPT-125M     | 4.049        | 0.811        | <b>0.812</b> | 0.327        |
| PTB          | GPT-2        | 5.028        | 0.892        | 1.646        | 0.231        |
|              | GPT-2-Medium | <b>4.769</b> | <b>0.875</b> | 1.394        | <b>0.208</b> |
|              | OPT-125M     | 5.199        | 0.907        | <b>0.935</b> | 0.382        |

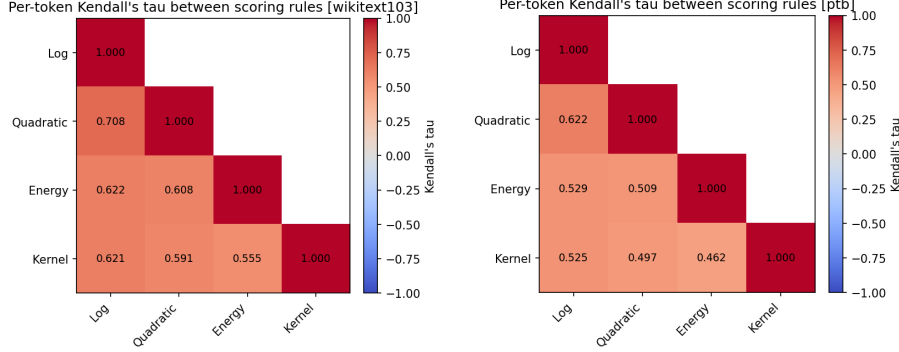


Figure 1: Experiment 1: token-level Kendall's  $\tau$  (defined in Appendix B) between per-token scores under the four rules, pooled across models. Left: WikiText-103; right: PTB. All pairs lie well below  $\tau = 1$ .

#### 4 Experiment 1: global rankings under the four rules

Table 2 reports the principal comparison. Under the log, quadratic, and kernel scores the verdict is unanimous on both corpora: GPT-2-Medium  $\succ$  GPT-2  $\succ$  OPT-125M. The energy score alone inverts this ordering, ranking OPT-125M first by a substantial margin (0.812 against 1.223 on WikiText-103) while the same model is ranked last by every other rule.

By Equation (2), each column of Table 2 estimates the mean Bregman divergence  $d_\varphi(\delta_y, P_t)$  between the realized outcomes and the models' forecasts under that rule's generator. The disagreement between columns is therefore a statement about geometry: OPT-125M's forecasts are farther from the data distribution than GPT-2-Medium's when distance is measured by KL divergence, squared Euclidean distance, or MMD, yet closer when measured by the energy distance taken in each model's own embedding space. The representation theorem does not merely permit such reversals; it implies they must occur whenever two models' error structures differ along directions that two generators curve differently, and Sections 5–8 identify those directions. Consistent with this reading, the rules also disagree pervasively at the level of individual tokens: pairwise Kendall's  $\tau$  between per-token scores lies between 0.46 and 0.71 across rule pairs and corpora (Figure 1).

#### 5 Experiment 2: tail behavior across token-frequency strata

Scored positions are stratified by the unigram-frequency decile of the target token (decile 0 containing the most frequent tokens). Figure 2 presents the resulting penalty profiles. Averaged across models on WikiText-103, the mean log-score increases by a factor of 2.05 between decile 0 and decile 8, against 1.22 for the quadratic score, 1.58 for the energy score, and 1.71 for the kernel score (corpus-level values in Appendix E).

The ordering of these amplification factors is determined by the boundary behavior of the generators in Table 1. For the log-score,  $d_\varphi(\delta_y, P) = -\log p_y$ , and the curvature  $1/p_v$  of the negative-entropy generator diverges as  $p_y \rightarrow 0$ ; since rare targets are precisely those to which models assign small probability, an unbounded penalty at the simplex boundary is a structural property of KL geometry,

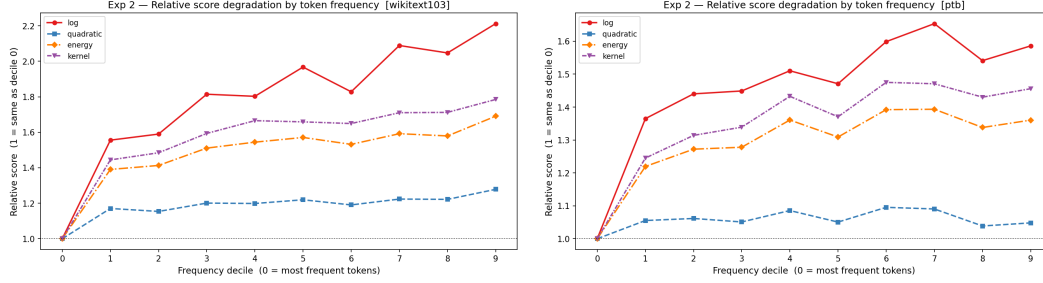


Figure 2: Experiment 2: mean score by unigram-frequency decile of the target token (left: WikiText-103; right: PTB). The qualitative pattern replicates across corpora.

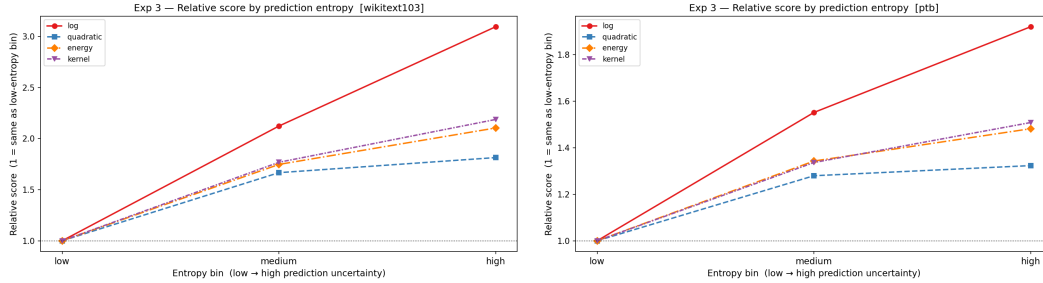


Figure 3: Experiment 3: mean score by predictive-entropy tertile (left: WikiText-103; right: PTB). The qualitative pattern replicates across corpora.

not an empirical accident. The quadratic generator’s constant Hessian bounds its divergence by 2, and the observed near-flat profile ( $1.22\times$ ) realizes that bound’s consequence. The energy and kernel scores occupy an intermediate position because their generators penalize rare-token errors through embedding distance rather than through  $p_y$ : the penalty grows only insofar as the predictive mass sits far from the target in feature space. The empirical ordering quadratic  $<$  energy  $<$  kernel  $<$  log of tail amplification is thus exactly the ordering of boundary curvatures of the four generators, as the representation theorem requires.

## 6 Experiment 3: score behavior under predictive uncertainty

Scored positions are next stratified into per-corpus tertiles of predictive entropy  $H(P) = -\sum_v p_v \log p_v$  (Figure 3). From the low to the high tertile on WikiText-103 the mean log-score increases by a factor of 3.09, the quadratic by 1.81, the energy by 2.10, and the kernel by 2.19. The more diagnostic statistic is the ratio between rules:  $S_{\log}/S_E$  rises from 2.58 in the low tertile to 3.79 in the high tertile, an increase of 47% (29% on PTB; Appendix F).

The widening ratio is a direct manifestation of locality. For the log-score,  $d_{\text{KL}}(\delta_y, P) = -\log p_y$  depends on the forecast only through the mass assigned to the realized token; a high-entropy forecast necessarily assigns smaller  $p_y$  and is penalized for the diffuseness itself, irrespective of where the remaining mass is placed. The energy divergence, by contrast, evaluates the full configuration of the forecast in embedding space: a diffuse  $P$  whose mass remains in the vicinity of  $e(y)$  can stand at a small energy distance from  $\delta_y$  even while  $-\log p_y$  is large. The monotone growth of the  $S_{\log}/S_E$  ratio across tertiles on both corpora is the empirical signature of these two response rates; the bounded, similarity-blind quadratic generator again sits between the extremes, completing the pattern the curvatures predict.

## 7 Experiment 4: statistical robustness of the ranking disagreement

To assess whether the disagreement of Section 4 could be a sampling artifact, we apply a nonparametric bootstrap over token positions: for each (model, corpus) cell the 20,000 cached per-token

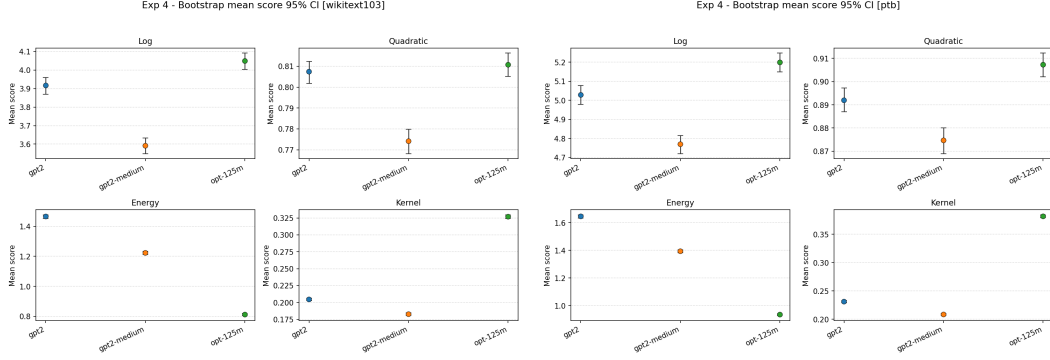


Figure 4: Experiment 4: bootstrap means and 95% percentile intervals ( $B = 1,000$ ) for every (model, rule) cell. Left: WikiText-103; right: PTB. Full tables appear in Appendix C.

scores are resampled with replacement  $B = 1,000$  times, and all rule means and induced rankings are recomputed on each replicate.

Three results emerge (Figure 4; Appendix C). First, the 95% percentile intervals around all twelve cell means of Table 2 are narrow relative to the gaps between models, so no pairwise comparison is ambiguous. Second, the joint event “OPT-125M is ranked first by the energy score and last by the log-score” obtains in 100% of replicates on both corpora. Third, the energy ranking agrees with the log-score ranking in 0% of replicates, the kernel in 100% (quadratic: 80% on WikiText-103, 100% on PTB).

Strict properness guarantees that each empirical rule mean is a consistent estimator of the corresponding expected Bregman divergence; the bootstrap addresses the remaining question of whether the *orderings* of those divergences are estimated stably at  $n = 20,000$ . The answer is affirmative: the divergence orderings induced by the four generators are statistically separated, so the cross-rule disagreement of Experiment 1 reflects a reproducible property of the models’ predictive distributions under the four geometries rather than noise. The bootstrap thereby licenses the mechanistic question taken up in Experiment 5—*which* structural property of the distributions the energy geometry is registering.

## 8 Experiment 5: an embedding-space account of the energy ranking

The energy divergence between a forecast and a point mass measures, by construction, how far the forecast’s probability mass lies from the realized token in embedding space. Its first-moment summary is the *predictive barycenter*  $c = \mathbb{E}_{X \sim P}[e(X)] = P^\top E$ : by Jensen’s inequality the leading term of Equation (3) satisfies  $\mathbb{E}\|e(X) - e(y)\| \geq \|c - e(y)\|$ , so the barycenter distance  $\|c - e(y)\|$  is a lower envelope of the energy penalty and a natural per-event diagnostic of what the energy score responds to. If the Bregman account of Experiment 1 is correct, the model ranked best by the energy rule should be the model whose barycenters lie closest to the realized tokens—most consequentially on *misprediction events* (64–78% of positions depending on cell), where the log and energy geometries render their most divergent verdicts.

Figure 5 confirms the prediction. On WikiText-103 misprediction events the mean barycenter distance is 1.44 for OPT-125M, 2.30 for GPT-2-Medium, and 2.63 for GPT-2 (PTB: 1.49, 2.34, 2.69)—exactly the energy-score ordering of Table 2, on both corpora; the same ordering holds for the distance from the modal (top-1) token to the target (Appendix D).

Experiment 5 closes the explanatory chain opened by Experiment 1. The energy generator’s divergence is small exactly when the forecast’s mass configuration sits near the target in feature space; the barycenter distance isolates that quantity event by event, and the model ordering it produces is the energy ordering—computed without reference to the score itself. The log-score, whose divergence  $-\log p_y$  is invariant to the embedding locations of the off-target mass, has no access to this structure and accordingly ranks the same events differently. Each rule is thus observed to be “satisfied” in precisely the sense the representation theorem specifies: its empirical verdict tracks its own generator’s

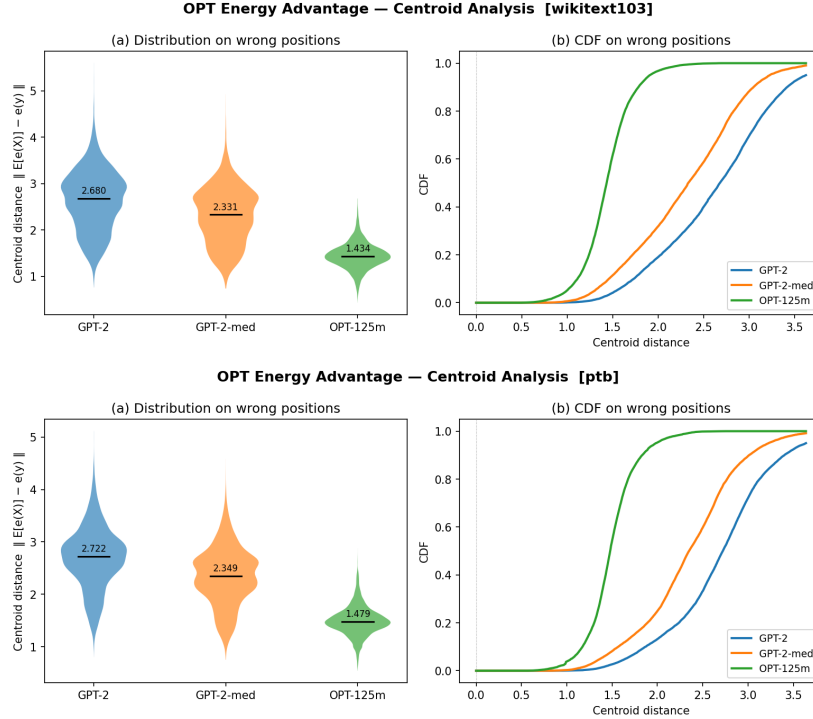


Figure 5: Experiment 5: barycenter-to-target distance  $\|\mathbb{E}_P[e(X)] - e(y)\|$  on misprediction events (top row: WikiText-103; bottom row: PTB). Within each row: distribution by model (black bar denotes the median) and empirical CDF. On both corpora the ordering of the three distributions reproduces the energy-score ranking of Table 2.

divergence, evaluated on the structural feature of the forecasts that the generator curves against. The corresponding scope restriction is taken up in Section 9.

## 9 Discussion

**Rankings are rule-relative, as the representation theorem requires.** Reporting a single rule reports a single geometry, and the choice of that geometry is substantive, not conventional: the tail and entropy sensitivities of Experiments 2–3 are properties of the log-score’s generator, not of the models. Neither is a defect, but evaluations that care about calibration on common tokens, or that do not wish to penalize acknowledged ambiguity, will reach materially different conclusions from a bounded rule obtained at negligible additional computational cost.

**What embedding-space scoring contributes.** The energy and kernel scores are the only rules in the suite whose generators are defined through a metric on tokens, and they extend evaluation in three directions: they award partial credit for semantically structured errors, which the log-score penalizes identically to arbitrary ones; they retain discriminative range where the log-score is dominated by position difficulty (rare targets, high-entropy contexts; Experiments 2–3); and the kernel bandwidth supplies a resolution parameter, with small  $\sigma$  approaching exact-match evaluation and large  $\sigma$  the energy rule’s diffuse assessment, so one strictly proper family spans several semantic scales.

**Scope of cross-model embedding comparisons.** The representation theorem also delimits what embedding-based comparisons can claim: when each model supplies its own embedding matrix the resulting divergences are defined over different metric spaces, and cross-model differences mix properties of the forecasts with properties of the representations. Within-model comparisons hold the geometry fixed and are free of this caveat; cross-model comparisons are sharpest when the feature map is fixed by design, e.g. a common reference embedding space.

**Future directions.** Two extensions appear most promising. The first is the systematic study of reference embedding spaces for cross-model comparison, as noted above. The second moves the energy score from evaluation to estimation. In the regression setting, Shen and Meinshausen (2024) introduce *engression*, in which a generative model of  $Y \mid X$  is fitted by minimizing the expected energy score; strict properness guarantees that the population minimizer is the true conditional distribution. Next-token prediction is formally the same problem, suggesting that language models could be fitted or fine-tuned against the energy score in embedding space, in direct analogy to engression—minimizing the energy divergence rather than the KL divergence of maximum likelihood, with gradients that do not diverge on rare targets and that reward semantically structured uncertainty (Experiments 2–3). Whether this improves calibration at scale is an open question this framework is positioned to test.

**Limitations.** The study covers three sub-500M-parameter models and two English corpora; the geometric mechanisms documented here are properties of the scoring rules and should transfer, but the specific rankings need not. The energy and kernel scores are Monte Carlo estimates ( $m = 200$ ) with negligible standard errors (Appendix C). A frequency-ranked CRPS variant was computed but excluded, as the total order it requires is arbitrary for tokens.

## 10 Conclusion

Perplexity evaluates a high-dimensional predictive object through a single Bregman geometry. Scoring identical predictions under four strictly proper rules shows that the geometries mostly concur, that their one persistent disagreement is statistically robust, and that every observed pattern is the one the Savage–Bregman representation requires. Evaluations should accordingly report at least one bounded rule and, where token similarity matters, one embedding-aware rule alongside perplexity. Proper scoring rules disagree only where models genuinely differ; reading those disagreements through their generators converts a nuisance of evaluation into an instrument of analysis.

## References

- Gneiting, T. and Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378.
- Gretton, A., Borgwardt, K. M., Rasch, M. J., Schölkopf, B., and Smola, A. (2012). A kernel two-sample test. *Journal of Machine Learning Research*, 13:723–773.
- Jelinek, F., Mercer, R. L., Bahl, L. R., and Baker, J. K. (1977). Perplexity—a measure of the difficulty of speech recognition tasks. *Journal of the Acoustical Society of America*, 62(S1):S63.
- Shen, X. and Meinshausen, N. (2024). Engression: Extrapolation through the lens of distributional regression. *Journal of the Royal Statistical Society, Series B*, 86(5):1361–1389.
- Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., and Sutskever, I. (2019). Language models are unsupervised multitask learners. OpenAI Technical Report.
- Steinwart, I. and Ziegel, J. F. (2021). Strictly proper kernel scores and characteristic kernels on compact spaces. *Applied and Computational Harmonic Analysis*, 51:510–542.
- Székel, G. J. and Rizzo, M. L. (2013). Energy statistics: A class of statistics based on distances. *Journal of Statistical Planning and Inference*, 143(8):1249–1272.
- Zhang, S., Roller, S., Goyal, N., et al. (2022). OPT: Open pre-trained transformer language models. arXiv:2205.01068.

## A Implementation details

**Pipeline.** Each (model, corpus) cell is processed in a single autoregressive pass: documents are tokenized (maximum length 512, special tokens added), logits are computed for positions  $1, \dots, T-1$ , and all rules are evaluated on the softmax of the same logits, so every rule scores the identical prediction events. Scoring terminates once 20,000 positions are accumulated per cell. All analyses are deterministic given the cached per-token arrays; analysis scripts fix the random seed at 591.



**Monte Carlo estimators.** For the energy score we use the paired-sample form: with  $X_1, \dots, X_m$  and  $X'_1, \dots, X'_m$  drawn independently from  $P$  ( $m = 200$ ),

$$\hat{S}_E = \frac{1}{m} \sum_{i=1}^m \|e(X_i) - e(y)\| - \frac{1}{2m} \sum_{i=1}^m \|e(X_i) - e(X'_i)\|,$$

which is unbiased for Equation (3); the kernel score uses the analogous plug-in for Equation (4) with  $k(e(y), e(y)) = 1$ . No  $T \times V$  intermediate matrices are materialized, keeping memory  $O(Tmd)$ .

**Bandwidths.** The median heuristic estimates  $\sigma_0$  from 20,000 random embedding pairs drawn from a 2,000-row subsample of each model’s embedding matrix.

**Compute.** All experiments run on CPU (Windows 11, no GPU); the full suite (Experiments 1–5, both corpora, three models) completes in approximately six hours, dominated by the  $3 \times 2 \times 20,000$  forward passes. Code, configuration, and per-token score caches are included in the project repository.

## B Token-level Kendall’s $\tau$ between rules

**Definition 1** (Kendall’s rank correlation coefficient). *Let  $(x_1, y_1), \dots, (x_n, y_n)$  be paired observations—here, the scores assigned to the same  $n$  token positions by two scoring rules. A pair of indices  $(i, j)$ ,  $i < j$ , is concordant if the two rules order the positions identically, i.e. if  $(x_i - x_j)(y_i - y_j) > 0$ , and discordant if  $(x_i - x_j)(y_i - y_j) < 0$ . Kendall’s coefficient is*

$$\tau = \frac{n_c - n_d}{\binom{n}{2}},$$

where  $n_c$  and  $n_d$  count concordant and discordant pairs. Thus  $\tau = 1$  indicates that the two rules induce identical orderings of the prediction events,  $\tau = -1$  exactly reversed orderings, and  $\tau = 0$  orderings that agree no more often than chance. In the presence of ties we report the standard tie-adjusted variant  $\tau_b$ , which replaces  $\binom{n}{2}$  by the geometric mean of the numbers of untied pairs in each variable (as implemented in SciPy). Because  $\tau$  depends on the scores only through their ranks, it is invariant to any monotone transformation of either rule and is therefore a natural measure of agreement between scoring rules whose numerical scales are not comparable.

Table 3: Pairwise Kendall’s  $\tau$  between per-token scores, pooled across models (10,000-position subsample per corpus; all  $p < 10^{-6}$ ).

| Rule pair          | WikiText-103 | PTB   |
|--------------------|--------------|-------|
| Log – Quadratic    | 0.708        | 0.622 |
| Log – Energy       | 0.622        | 0.529 |
| Log – Kernel       | 0.621        | 0.526 |
| Quadratic – Energy | 0.608        | 0.509 |
| Quadratic – Kernel | 0.591        | 0.497 |
| Energy – Kernel    | 0.555        | 0.462 |

## C Bootstrap tables

Event probabilities across replicates (identical on both corpora):  $\Pr[\text{OPT-125M ranked first under energy}] = 1.00$ ;  $\Pr[\text{OPT-125M ranked last under log}] = 1.00$ ;  $\Pr[\text{both events jointly}] = 1.00$ . Fraction of replicates whose full three-model ranking matches the log-score ranking: energy 0.00, kernel 1.00, quadratic 0.80 on WikiText-103; energy 0.00, kernel 1.00, quadratic 1.00 on PTB.

## D Experiment 5 summary statistics

## E Experiment 2 amplification factors

## F Experiment 3 entropy-tertile ratios

Table 4: Bootstrap ( $B = 1,000$  resamples of 20,000 token positions) standard errors and 95% percentile intervals for every cell of Table 2, WikiText-103. All cross-model gaps exceed ten standard errors.

| Model        | Rule      | Mean  | SE    | CI <sub>2.5</sub> | CI <sub>97.5</sub> |
|--------------|-----------|-------|-------|-------------------|--------------------|
| GPT-2        | Log       | 3.917 | 0.024 | 3.870             | 3.964              |
| GPT-2        | Quadratic | 0.807 | 0.003 | 0.801             | 0.813              |
| GPT-2        | Energy    | 1.466 | 0.006 | 1.453             | 1.477              |
| GPT-2        | Kernel    | 0.205 | 0.001 | 0.203             | 0.206              |
| GPT-2-Medium | Log       | 3.591 | 0.024 | 3.545             | 3.638              |
| GPT-2-Medium | Quadratic | 0.774 | 0.003 | 0.768             | 0.780              |
| GPT-2-Medium | Energy    | 1.223 | 0.005 | 1.212             | 1.233              |
| GPT-2-Medium | Kernel    | 0.183 | 0.001 | 0.181             | 0.185              |
| OPT-125M     | Log       | 4.049 | 0.025 | 4.001             | 4.097              |
| OPT-125M     | Quadratic | 0.811 | 0.003 | 0.805             | 0.816              |
| OPT-125M     | Energy    | 0.812 | 0.003 | 0.806             | 0.818              |
| OPT-125M     | Kernel    | 0.327 | 0.001 | 0.324             | 0.330              |

Table 5: Barycenter and modal-token distances to the target on misprediction events (mean over events; PTB in parentheses). The ordering of either column reproduces the energy-score ranking of Table 2 on both corpora.

| Model        | % positions mispredicted | Barycenter dist. | Top-1 dist. |
|--------------|--------------------------|------------------|-------------|
| GPT-2        | 67.9 (76.9)              | 2.63 (2.69)      | 3.50 (3.51) |
| GPT-2-Medium | 64.3 (74.6)              | 2.30 (2.34)      | 3.03 (3.01) |
| OPT-125M     | 68.7 (78.2)              | 1.44 (1.49)      | 2.01 (2.04) |

Table 6: Ratio of mean score at frequency decile 8 (rare) to decile 0 (common), averaged across models.

| Corpus       | Log   | Quadratic | Energy | Kernel |
|--------------|-------|-----------|--------|--------|
| WikiText-103 | 2.05× | 1.22×     | 1.58×  | 1.71×  |
| PTB          | 1.54× | 1.04×     | 1.34×  | 1.43×  |

Table 7: Mean log-to-energy and log-to-kernel score ratios by predictive-entropy tertile, averaged across models.

| Corpus       | Tertile | Log/Energy | Log/Kernel |
|--------------|---------|------------|------------|
| WikiText-103 | low     | 2.58       | 12.9       |
|              | medium  | 3.13       | 15.5       |
|              | high    | 3.79       | 18.2       |
| PTB          | low     | 3.23       | 15.7       |
|              | medium  | 3.73       | 18.2       |
|              | high    | 4.18       | 20.0       |